

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.

3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Reference Resolution

Algorithm

1. Take the input discourse as a sequence of sentences.
2. Identify all pronouns such as he, she, it, they, his, her, etc.
3. Identify possible antecedents (entities) mentioned earlier in the discourse.
4. Check gender and number agreement between the pronoun and candidate antecedents.
5. Apply recency, giving preference to a recent compatible entity.
6. Check semantic compatibility between the pronoun and the candidate.
7. Select the most appropriate antecedent.
8. Replace the pronoun with its antecedent.
9. Generate the resolved discourse.

Given Text

"Ravi met Arun at the library. He borrowed a book and later returned it."

Reference Resolution

Pronoun	Possible Antecedents	Resolved Entity	Reason
He	Ravi, Arun	Ravi	Both are male, but Ravi is the main discourse subject.
it	a book	a book	"It" is singular and refers to the object that was borrowed.

Resolved Discourse

"Ravi met Arun at the library. Ravi borrowed a book and later returned the book."

Validation

The discourse context is important for resolving the pronouns. Both Ravi and Arun are possible male antecedents for "He." However, Ravi is the subject of the previous sentence and is therefore preferred because of subject continuity and discourse prominence.

The pronoun "it" clearly refers to "a book" because the book is the object that can naturally be borrowed and returned.

Resolution Flow

Ravi + Arun

↓

Identify "He"

↓

Gender/Number Agreement

↓

Discourse Subject Preference



"He" → Ravi

Identify "it"



Semantic Compatibility



"it" → book

2. Text Coherence and Discourse Structure

Algorithm

1. Divide the input text into individual sentences.
2. Identify the main events or propositions in each sentence.
3. Identify discourse markers such as **therefore, because, then, after, however**, etc.
4. Compare consecutive sentences.
5. Identify logical relations such as:
 - Cause–Effect
 - Sequence
 - Elaboration
 - Contrast
 - Condition
6. Construct a discourse structure showing how the sentences are connected.
7. Evaluate whether the relationships create a logical and coherent text.

Given Discourse

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Sentence Analysis

Sentence	Main Information
S1	Heavy rainfall caused the roads to become flooded.
S2	Because of the situation, schools were closed.
S3	Students attended classes online as a consequence.

Discourse Relations

Sentences	Relation	Explanation
S1 → S2	Cause–Effect	Flooded roads resulted in schools being closed.
S2 → S3	Cause–Effect / Consequence	School closure led students to attend classes online.
S1 → S3	Overall Consequence	Heavy rainfall indirectly resulted in online classes.

Discourse Structure Representation

Heavy Rainfall



Roads Were Flooded



Cause–Effect



Schools Were Closed



Consequence



Students Attended

Classes Online

Coherence Explanation

The text is coherent because every sentence is logically connected to the previous sentence.

The relationship can be represented as:

Heavy rainfall



Flooded roads



School closure



Online classes

The word "Therefore" explicitly indicates a cause-and-effect relationship between the first and second sentences. The third sentence provides the consequence of school closure.

Thus, the sentences form a meaningful sequence rather than unrelated statements.

3. Conversational Agent and Dialogue Acts

Algorithm

1. Receive each user or agent utterance.
2. Analyze the words and sentence structure.
3. Identify the speaker's intention.
4. Classify the utterance into a dialogue act:
 - Request
 - Question
 - Inform
 - Confirmation
 - Action
5. Maintain the conversation context.
6. Use the identified dialogue act to determine the next appropriate response.
7. Continue until the user's task is completed.

Given Conversation

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Dialogue Act Classification

Speaker	Utterance	Dialogue Act	Explanation
User	"Can you book a train ticket for me?"	Request	The user requests the agent to book a ticket.
Agent	"Sure, where would you like to travel?"	Question	The agent asks for required information.
User	"I want to go to Chennai."	Inform	The user provides the destination.
Agent	"Your ticket has been booked."	Confirmation	The agent confirms that the requested task is completed.

Dialogue-Act Sequence

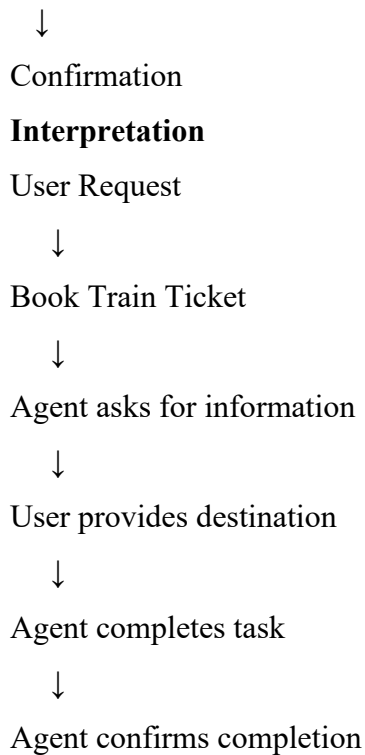
Request

↓

Question

↓

Inform



Evaluation

Dialogue acts help the conversational agent understand what the user wants and what information is required.

- Request identifies the user's goal.
- Question allows the agent to collect missing information.
- Inform provides the required information.
- Confirmation tells the user that the task has been completed.

Therefore, dialogue-act recognition enables the conversational agent to maintain context and respond appropriately.

4. Language Generation and Surface Realization

Algorithm

1. Receive the semantic representation.
2. Identify the **Agent**, **Action**, **Object**, and **Tense**.
3. Select appropriate words from the vocabulary.
4. Arrange the words according to English grammar.
5. Convert the action into the required tense.
6. Add determiners such as **the** or **a** where required.
7. Perform surface realization.
8. Check grammatical correctness.

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

Step 1: Lexical Selection

The semantic concepts are converted into English words:

Agent → student

Action → buy

Object → book

Determiners are added:

student → the student

book → a book

Step 2: Tense Processing

The input specifies **Past Tense**.

Therefore:

buy → bought

Step 3: Sentence Structuring

English normally follows:

Subject + Verb + Object

Therefore:

The student + bought + a book

Final Sentence

"The student bought a book."

Surface Realization Process

Semantic Representation



Lexical Selection



"The student" + "buy" + "a book"



Tense Processing



"The student" + "bought" + "a book"



Surface Realization



The student bought a book.

Grammatical Validation

The generated sentence is grammatically correct because:

1. "The student" is a valid singular subject.
2. "Bought" is the correct past-tense form of "buy."
3. "A book" is the correct object phrase.
4. The sentence follows the standard Subject–Verb–Object (SVO) structure.

Therefore, the final generated sentence is:

The student bought a book.

5. Interlingua-Based and Statistical Machine Translation

Algorithm

1. Accept the source-language sentence.
2. Perform lexical, syntactic, and semantic analysis.
3. Identify the subject, action, object, tense, and aspect.
4. Convert the meaning into an Interlingua representation.
5. Generate several possible target-language translations.
6. Calculate statistical scores for the candidate translations.
7. Use translation probability and language-model probability.
8. Select the candidate with the highest score.
9. Perform final surface realization.
10. Output the best translation.

Source Sentence

"The boy is playing football."

Step 1: Analyze the Source Sentence

The sentence contains:

Word/Phrase Role

The boy	Subject / Agent
is playing	Verb / Action
football	Object
is playing	Present Progressive Tense

Semantic information:

Agent = Boy

Action = Play

Object = Football

Tense = Present

Aspect = Progressive

Step 2: Create Interlingua Representation

Interlingua represents the meaning independently of a particular language.

PLAY

|— Agent: BOY

|— Object: FOOTBALL

|— Tense: PRESENT

|— Aspect: PROGRESSIVE

Or:

PLAY(

Agent = BOY,

Object = FOOTBALL,

Tense = PRESENT,

Aspect = PROGRESSIVE

)

This representation captures the meaning of the original sentence rather than simply copying its English word order.

Step 3: Generate Candidate Target-Language Translations

Assuming the target language is Tamil, possible candidate translations can be generated from the Interlingua representation.

Candidate 1:

The boy is playing football.

Candidate 2:

The boy plays football.

Candidate 3:

The boy is playing the football.

For an actual statistical translation system, these candidates would be generated in the target language and scored according to the learned translation and language models.

Step 4: Apply Statistical Scoring

A statistical translation system evaluates each candidate using probabilities.

A simplified scoring function is:

Score(Target) =

Translation Probability

×

Language Model Probability

Example:

Candidate	Translation Probability	Language Fluency	Overall Result
Candidate 1	High	High	Best
Candidate 2	Medium	High	Good
Candidate 3	Low	Medium	Less suitable

The system selects the candidate with the highest overall probability.

Step 5: Final Translation

For the English target-language realization, the selected sentence is:

The boy is playing football.

If translating into another language, the selected target-language candidate would undergo final grammatical and morphological processing before being produced.

Overall Translation Architecture

Source Sentence



Source Language Analysis



Semantic Analysis



Interlingua Representation



Candidate Translation Generation



Statistical Scoring



Translation Probability

+

Language Model Probability



Best Candidate Selection



Surface Realization



Final Translation

Evaluation

Advantages of Interlingua

- Represents the meaning of the source sentence.
- Reduces dependence on source-language word order.
- Separates semantic analysis from target-language generation.
- Helps preserve important information such as agent, action, object, tense, and aspect.
- Can reduce ambiguity during translation.

Advantages of Statistical Translation

- Compares multiple possible translations.
- Uses probabilities learned from translation data.
- Helps select fluent and natural target-language sentences.
- Handles alternative word choices and grammatical structures.
- Reduces the possibility of selecting an unlikely translation.

Combined Approach

Interlingua

+

Statistical Scoring



Better Meaning Preservation

+

Better Fluency

+

Reduced Ambiguity



Improved Translation Quality

Thus, combining Interlingua-based semantic representation with statistical translation methods provides both meaning preservation and natural target-language generation.